

KLEINE BROGEL AB, Belgium

WMO: 064790

Lat: 51.1689N

Lon: 5.4628E

Elev: 200

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	22.9	13.1	10.8	22.6	17.6	13.4	25.7	26.5	49.5	23.7	49.4	3.9	70	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	87.6	68.3	83.5	66.5	79.8	65.1	70.3	82.4	68.5	79.9	66.8	76.6	6.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	98.1	74.2	64.7	92.1	72.1	63.1	87.0	70.5	34.3	82.4	32.8	80.2	31.4	76.9	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	17.9	15.5	DB	12.2	95.3	6.2	3.9	7.7	98.1	4.1	100.3	0.6	102.5	-3.9	105.3
			WB	11.7	74.2	6.1	2.1	7.3	75.8	3.8	77.0	0.4	78.2	-4.1	79.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	37.7	39.2	44.0	50.3	56.8	62.1	65.9	64.9	59.2	52.3	44.4	39.0	(6)
	DBStd	11.91	7.82	7.28	6.48	6.77	6.31	6.01	5.94	5.29	5.16	6.28	6.64	7.50	(7)
	HDD50	1571	382	307	204	79	13	0	0	0	1	48	190	347	(8)
	HDD65	5191	845	724	652	443	265	127	61	67	188	393	619	807	(9)
	CDD50	2071	2	3	18	88	222	363	493	460	276	120	21	5	(10)
	CDD65	215	0	0	0	1	9	40	89	62	13	1	0	0	(11)
	CDH74	2460	0	0	0	50	182	480	932	672	136	8	0	0	(12)
	CDH80	784	0	0	0	7	37	140	331	241	28	0	0	0	(13)
(14) Wind	WSAvg	6.6	7.9	7.6	7.4	6.3	6.3	6.1	5.9	5.4	5.6	6.2	6.5	7.7	(14)
(15) Precipitation	PrecAvg	32.2	2.7	2.6	2.1	1.9	2.3	3.0	3.1	3.1	2.4	2.6	2.9	3.3	(15)
	PrecMax	41.5	5.6	6.1	4.6	3.9	4.6	8.9	6.7	7.0	6.9	7.3	5.1	6.1	(16)
	PrecMin	24.9	0.0	0.4	0.0	0.1	0.8	0.6	0.8	0.6	0.6	1.0	0.5	1.1	(17)
	PrecStd	4.0	1.3	1.3	1.1	0.9	1.0	1.6	1.5	1.8	1.6	1.2	1.1	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	59.0	68.5	80.3	84.7	90.1	94.3	92.5	83.8	74.2	63.2	56.8	(19)
		MCWB	51.4	49.5	53.7	59.6	64.5	68.6	69.6	69.8	66.4	63.1	56.8	52.7	(20)
	2%	DB	52.9	54.4	62.7	73.8	78.7	83.9	87.9	86.7	77.8	68.7	59.1	54.4	(21)
		MCWB	49.9	48.7	51.7	57.3	61.5	66.4	68.7	68.4	64.8	61.0	54.7	51.4	(22)
	5%	DB	50.3	51.7	58.0	68.3	74.6	79.5	83.2	81.6	72.9	64.8	56.5	52.1	(23)
		MCWB	47.7	47.7	50.0	55.1	59.9	64.7	66.7	66.9	62.8	59.1	53.0	49.5	(24)
	10%	DB	48.1	49.5	54.4	63.3	70.1	75.3	79.1	76.7	68.9	61.8	54.0	49.5	(25)
		MCWB	45.7	46.5	48.5	52.8	58.3	62.8	65.7	64.7	60.5	57.0	51.2	47.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	51.9	55.7	61.3	67.6	71.7	72.9	72.8	68.8	64.3	57.9	54.0	(27)
		MCDB	54.4	55.5	64.2	75.3	79.8	85.3	87.0	86.0	79.0	72.0	61.3	55.6	(28)
	2%	WB	50.3	50.1	53.4	58.5	63.9	68.5	70.6	69.9	66.2	61.9	55.4	51.9	(29)
		MCDB	52.2	53.0	59.5	70.0	74.4	79.8	83.0	81.5	74.7	67.2	58.4	54.1	(30)
	5%	WB	48.0	48.5	51.3	56.0	61.3	66.3	68.6	67.9	64.0	59.8	53.5	49.7	(31)
		MCDB	50.2	51.1	56.5	66.0	70.8	76.3	80.2	79.3	71.3	64.0	56.0	51.9	(32)
	10%	WB	45.8	46.5	49.3	53.9	59.4	64.2	66.7	66.1	61.9	57.9	51.3	47.4	(33)
		MCDB	47.9	49.4	53.9	61.8	68.0	73.0	76.2	74.8	67.5	61.3	53.7	49.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	11.7	15.9	20.3	20.0	20.4	20.0	20.1	18.0	14.9	11.0	8.9	(35)
		MCDBR	10.3	14.2	23.3	29.0	28.5	28.0	28.0	28.6	24.3	18.7	12.7	9.6	(36)
	5% WB	MCWBR	8.6	9.9	13.7	14.7	13.3	12.8	11.1	12.3	12.5	11.5	9.3	7.9	(37)
		MCWBR	9.9	11.6	19.0	25.4	23.8	24.1	24.4	24.7	21.1	17.0	11.6	9.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.329	0.353	0.385	0.417	0.424	0.422	0.429	0.419	0.391	0.367	0.342	0.326	(40)
		taud	2.380	2.341	2.255	2.199	2.222	2.244	2.228	2.277	2.345	2.398	2.384	2.349	(41)
	Ebn at Noon		217	242	255	260	264	265	261	257	251	234	208	196	(42)
		Edh at Noon	20	27	35	41	42	42	42	38	32	25	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		241	456	824	1318	1577	1710	1610	1374	1023	602	283	187	(44)
	RadStd		31	89	117	146	140	143	193	130	99	49	30	25	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (89 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page